

Astronomy 98 Final Project

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1 Background

Intensity is a measure of the amount of light produced per unit area. For galaxies, this value decreases exponentially due to the decrease in stellar density as you get further from the galactic bulge. Although all galaxies follow an exponential light profile, the slope of the exponential decay varies for different galaxy types. One way to model this is through the Sérsic profile given by

$$\mu(r) = \mu_e + \frac{2.5 \cdot b_n}{\ln(10)} \left[\left(\frac{r}{r_e} \right)^{1/n} - 1 \right]. \quad (1)$$

$$b_n \approx 2n - 0.324 \quad (2)$$

Where $\mu(r)$ is the intensity of the galaxy at any given radial distance in units of magnitudes, μ_e is the half-light intensity (or half the total intensity of the galaxy), b_n defines the shape of the light profile, r is the radial distance from the center of the galaxy, r_e is the effective radius that encloses half of the total light of the galaxy, and n is the Sérsic index. Common values for the Sérsic index are $n = 1$ for spiral galaxies and $n = 4$ for elliptical galaxies.

2 Tools/Resources

For this project, I used several Python packages to execute what I wanted to do. These included:

- Numpy - Used to create arrays and mathematical tools for proper data analysis
- Pandas - Used to organize and clean the initial data that I downloaded
- Matplotlib - Used for all of the plotting for this project
- Scipy - Used for the regression analysis used to find the Sérsic index of the galaxies

On top of just these packages, I also downloaded data from the Strasburg astronomical data center. Specifically I downloaded data from this [2] dataset.

3 Methods

3.1 Cleaning

The raw data I downloaded had several issues that I had to clean before I could do meaningful analysis with it. The first of these issues was the inclusion of empty cells in some of the data entries, so I first removed those blank entries. Then I had to separate the data into each of the galaxies to make the data more usable.

3.2 Fitting/Plotting

After cleaning the data, I then looped through all 48 galaxies and performed the fit to Eq (1) getting all relevant parameters, and then plotting the raw data and the fit for each galaxy on a massive subplot.

4 Results

5 Conclusion

References

- [1] Graham, A. (2005) *A CONCISE REFERENCE TO (PROJECTED) SÉRSIC $R1/n$ QUANTITIES, INCLUDING CONCENTRATION, PROFILE SLOPES, PETROSIAN INDICES, AND KRON MAGNITUDES.*
<https://ned.ipac.caltech.edu/level5/March05/Graham/Graham2.html#note2>

- [2] Bouquin, A. (2018) *The GALEX/ S^{4G} surface brightness and color profiles catalog. I. Surface photometry and color gradients of galaxies.* <https://cdsarc.cds.unistra.fr/viz-bin/cat/J/ApJS/234/18>